Reasoning techniques & case study

Dynamic Modelling for Human-centered Systems

Chapter 14b

Overview of today's lecture

1. **Summary previous lectures**
2. Reasoning techniques
3. Case study

Embedding domain models

- Analysis model
  - Find out whether some **undesired state** occurs
  - Based on domain model + desired state

- Support model
  - Propose a **support action** that will have a positive effect (on the undesirable state)
  - Based on domain set of possible **support actions**

Overview of today's lecture

1. Summary previous lecture
2. **Reasoning techniques**
3. Case study

Model-Based Reasoning

- Long history in AI
- First used in **medical diagnosis**
- Main question: given a numbers of **observations** about the patient (symptoms), can we determine which disease or **condition** explains these symptoms?

"I have no idea what's wrong with you. I just collect information. My computer makes the decisions."

Available modules

Symptoms & Signs Available:

CHEST PAIN
CHEST PAIN (Pro Version)
CONSTIPATION
COUGH
FAINTING (SYNCOPE)
FATIGUE
HEADACHE
BACK PAIN
VERTIGO (DIZZINESS)
UPPER ABDOMINAL SYMPTOMS

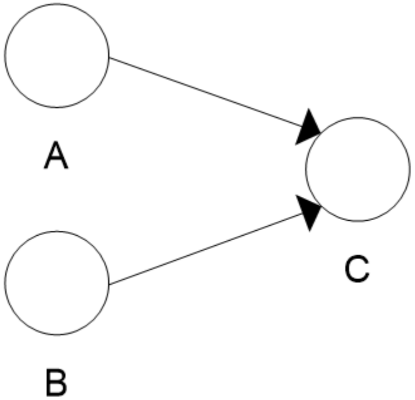
Conditions Available:

ALCOHOLISM
DEPRESSION
GALL BLADDER DISEASE
HEARING LOSS
SEIZURES AND EPILEPSY
SEIZURES/EPILEPSY (Pro Version)
ABDOMINAL CONDITIONS

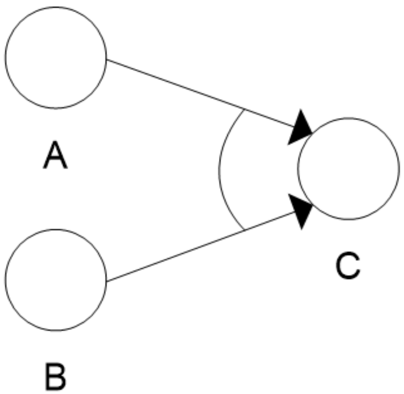
Model-Based Reasoning

- **Forward** in time
  - from causes to effects
  - (deduction)
- **Backward** in time
  - from effects to causes
  - (abduction)
- Different *variants* of causal relations
  - one-to-one
  - one-to-many
  - many-to-one
    - disjunction
    - conjunction

Refined visualization



- Disjunction (“OR”)
  - one of the inputs is sufficient to cause C



- Conjunction (“AND”)
  - both inputs are required to cause C

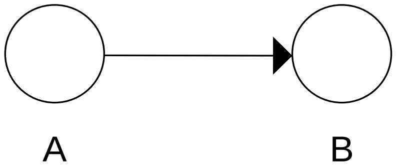
Reasoning methods in analysis & support models

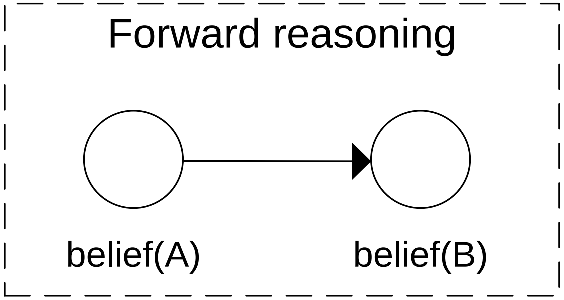
- When using a domain model as basis for analysis model, or support model we need to **transform** our domain model
  - for analysis model: it depends on the order of the observations and desired state
  - for support model: there are two approaches

- Question: **which beliefs can we generate** in our analysis / support model?
  - when applying forward reasoning or backward reasoning

Forward Reasoning (1)

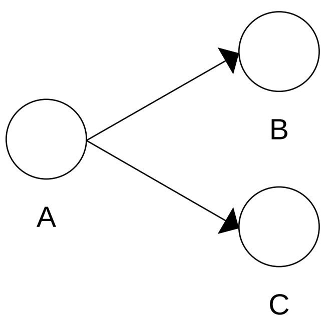
- One cause, one effect

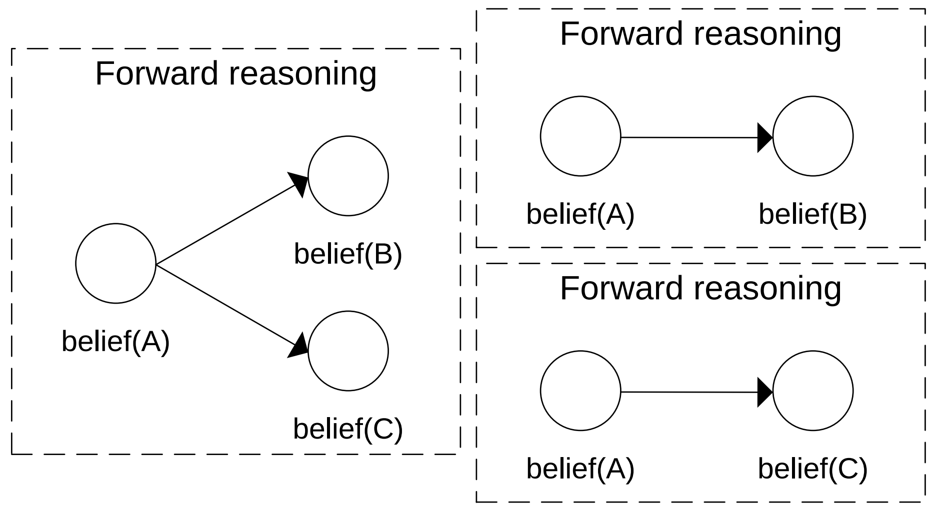




Forward Reasoning (2)

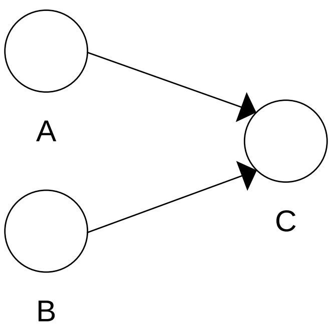
- One cause, two effects

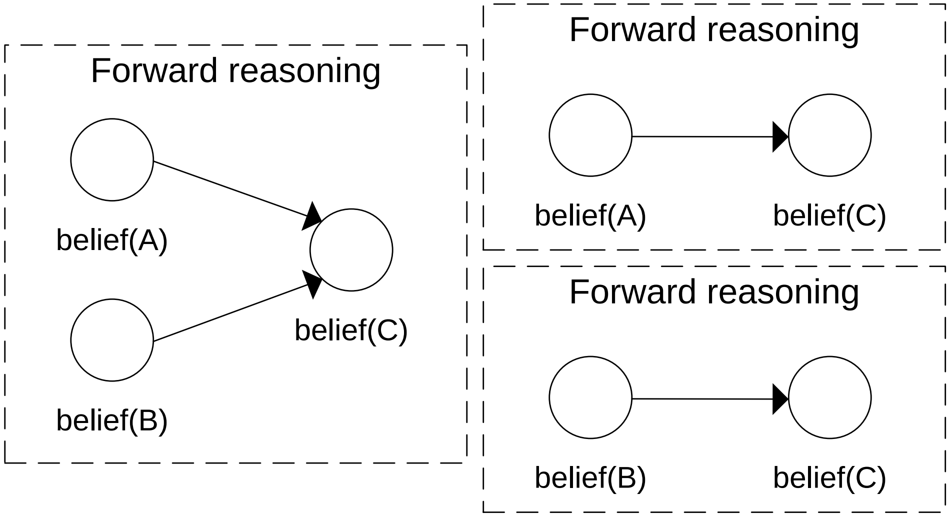




Forward Reasoning (3)

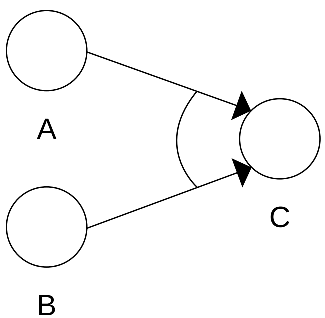
- Two disjoint causes

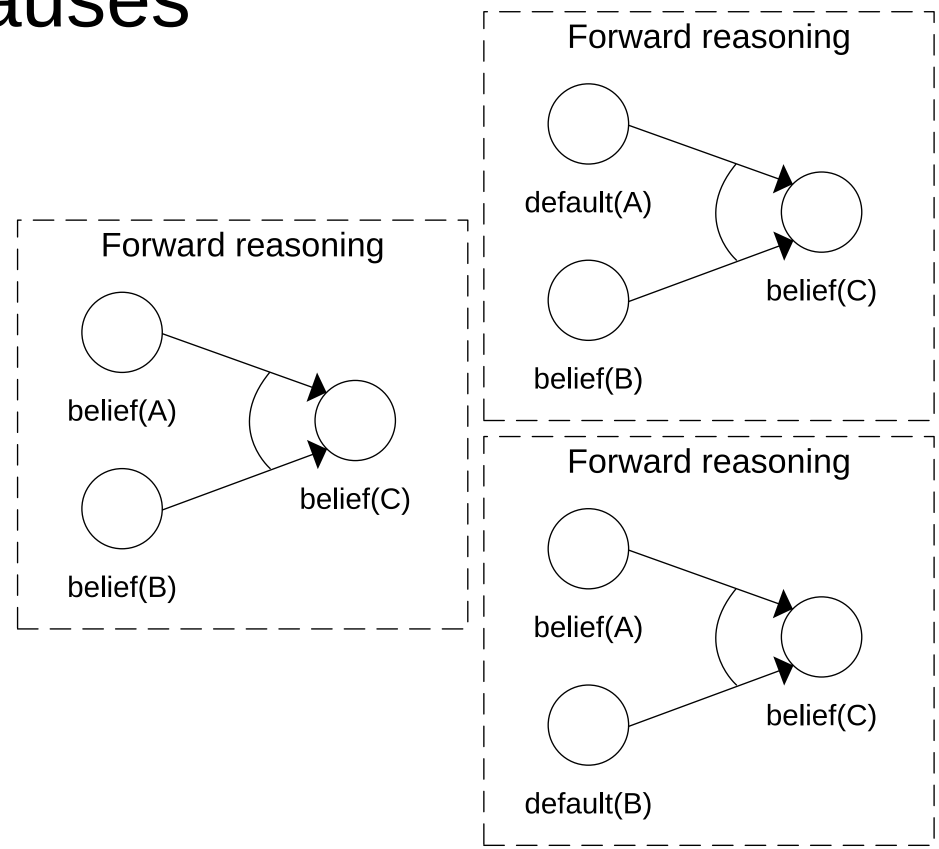




Forward Reasoning (4)

- Two combined causes



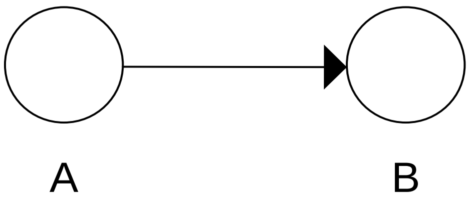


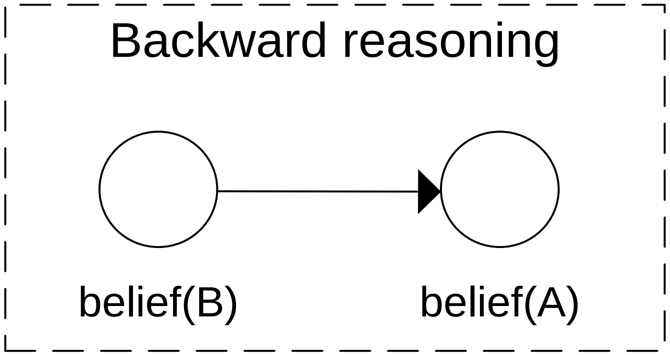
Backward Reasoning

- Logical fallacy: inverting the conditional statement
- Abduction: finding plausible solution

Backward Reasoning (1)

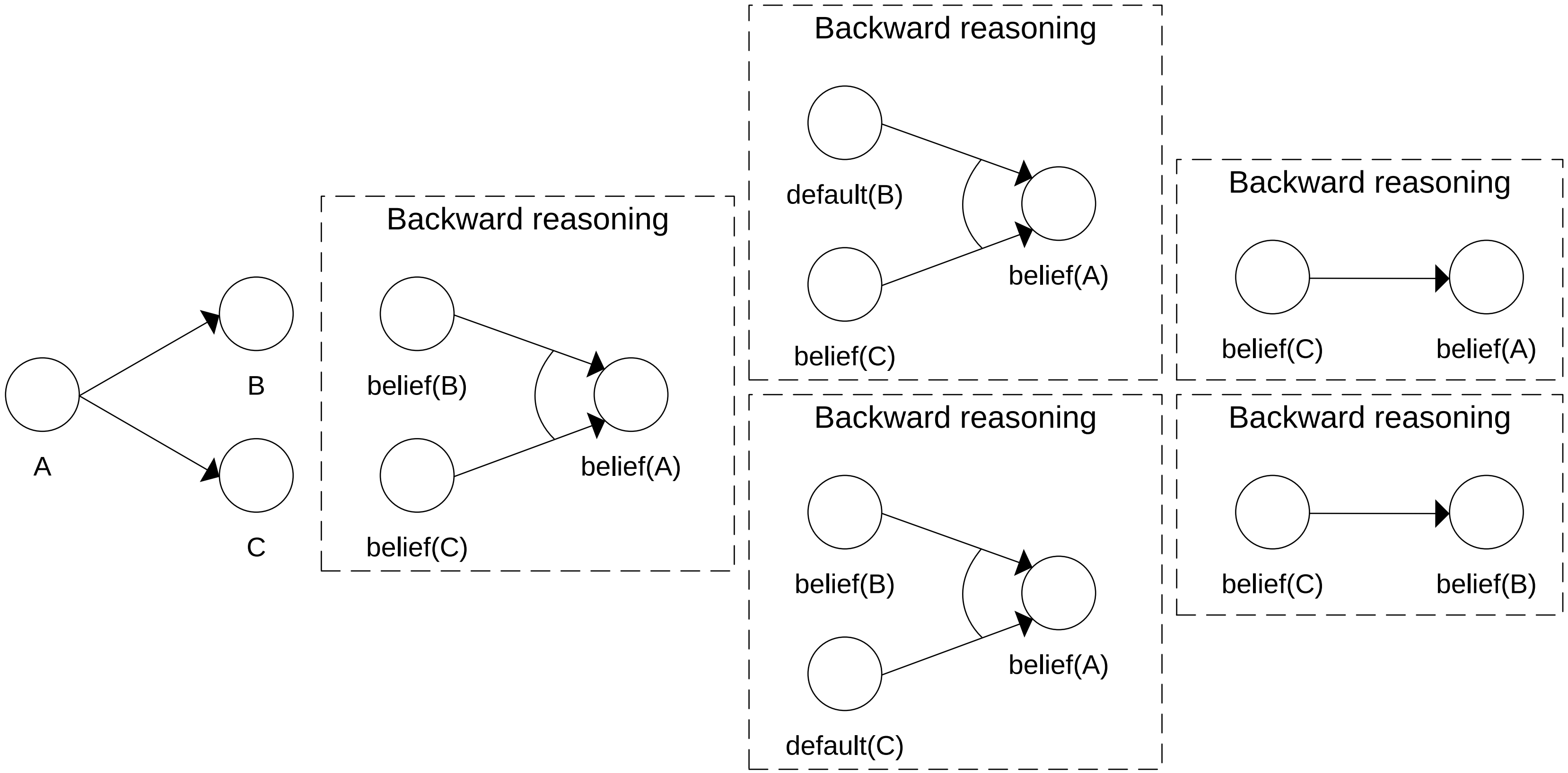
- One cause, one effect





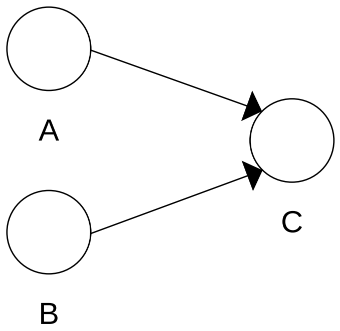
Backward Reasoning (2)

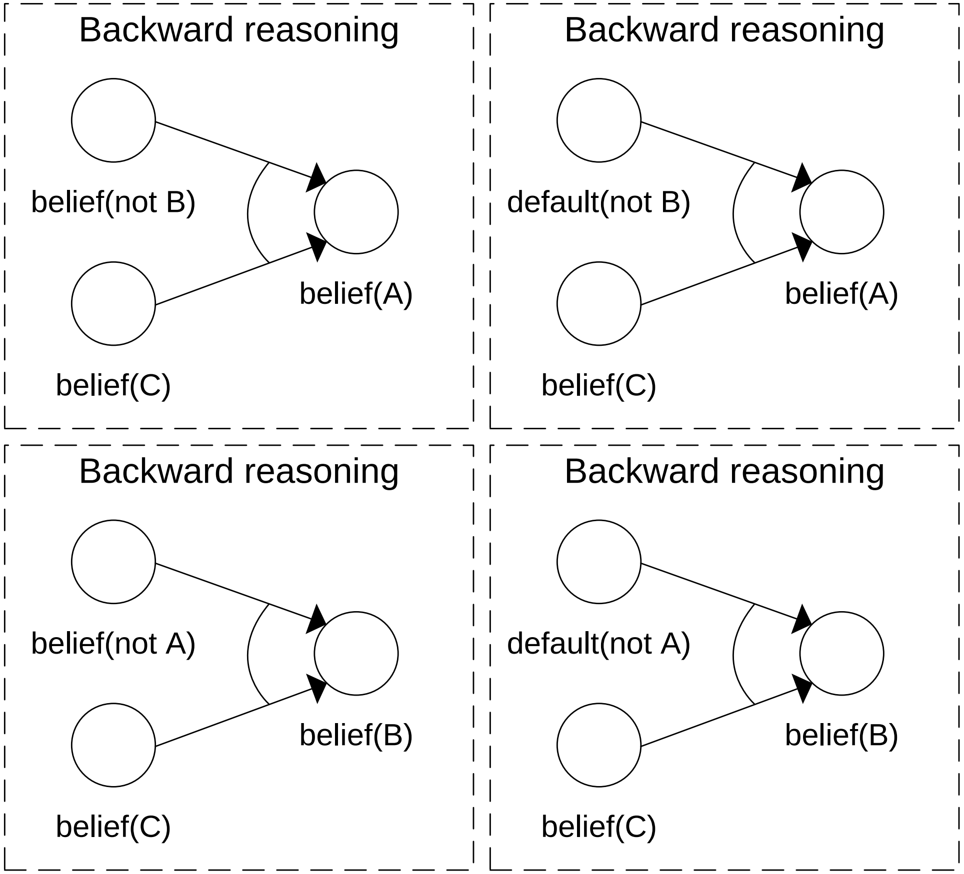
- One cause, two effects



Backward Reasoning (3)

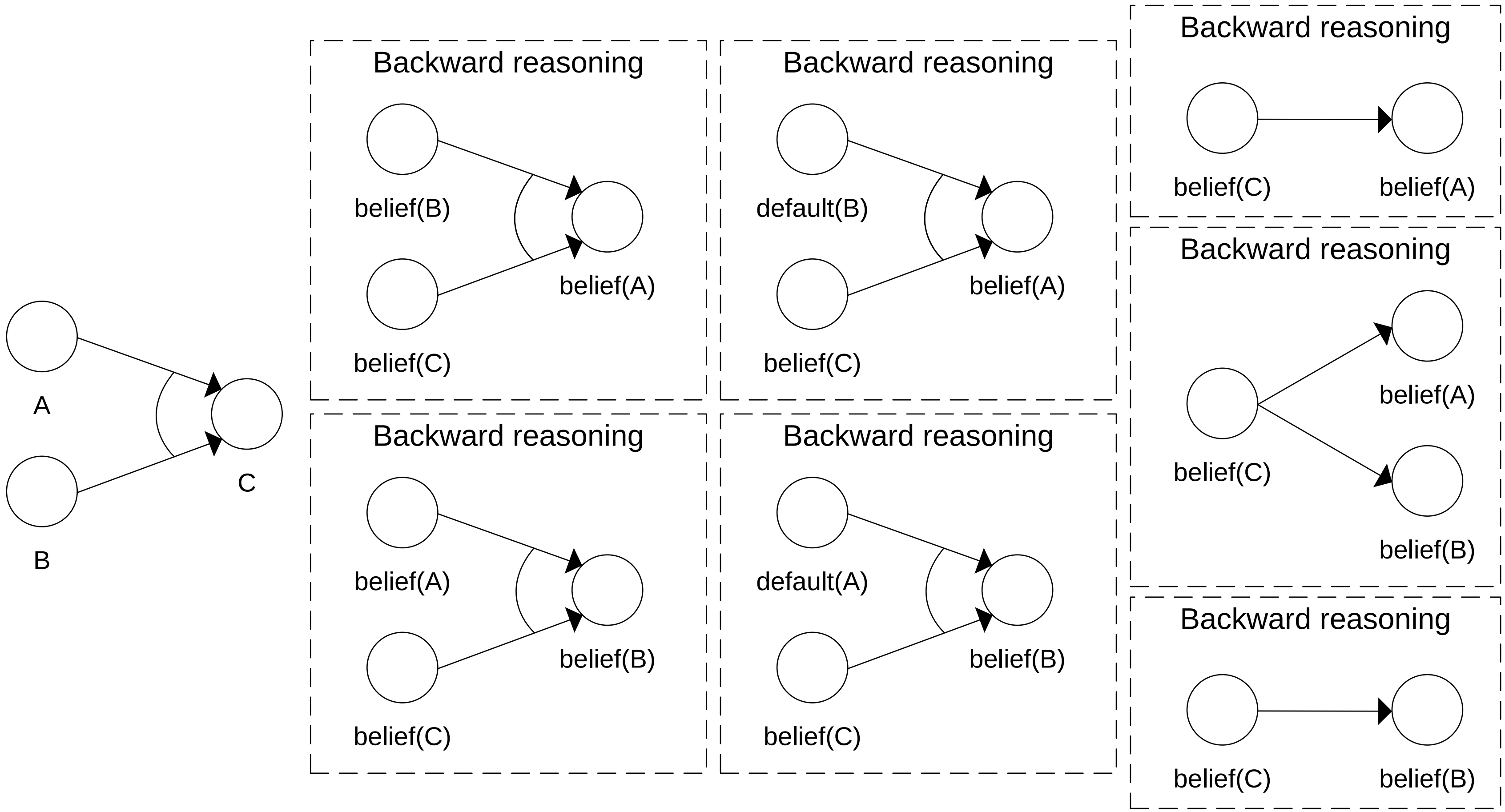
- Two disjoint causes





Backward Reasoning (4)

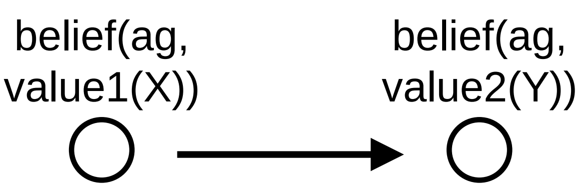
- Two combined causes



Numerical Values

IF value1(X)
THEN value2(X*0,5)

DOMAIN



IF belief(ag, value1(X))
THEN belief(ag, value2(X*0,5))

FORWARD

Numerical Values

value1(X) value2(Y)

IF value1(X)
THEN value2(X*0,5)

DOMAIN

belief(ag, value2(Y)) belief(ag, value1(X))

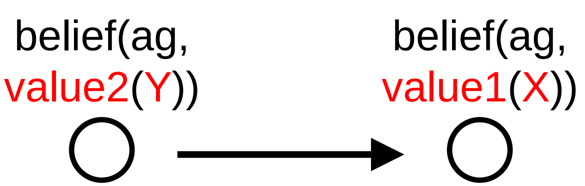
IF belief(ag, value2(Y))
THEN belief(ag, value1(Y*2))

BACKWARD

Numerical Values

IF value1(X)
THEN value2(X+0,5(1-X))

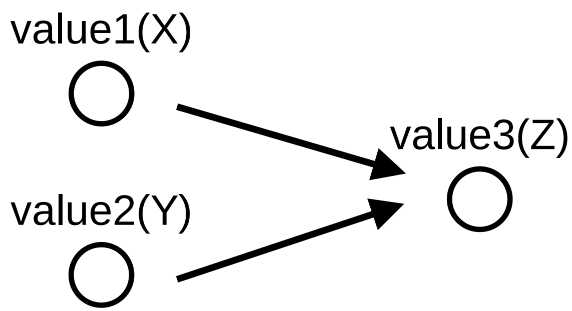
DOMAIN



IF belief(ag, value2(Y))
THEN belief(ag, value1(2*Y-1))

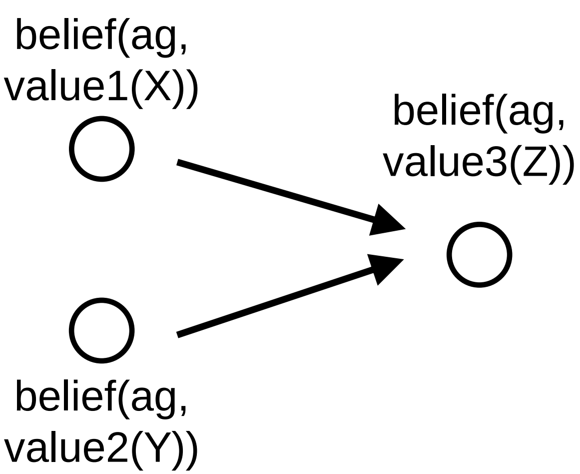
BACKWARD

Numerical Values: Many-to-one



IF value1(X)
AND value2(Y)
THEN value3(X+Y)

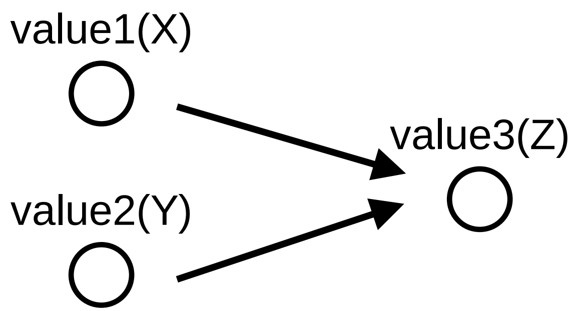
DOMAIN



IF belief(ag, value1(X))
AND belief(ag, value2(Y))
THEN belief(ag, value3(X+Y))

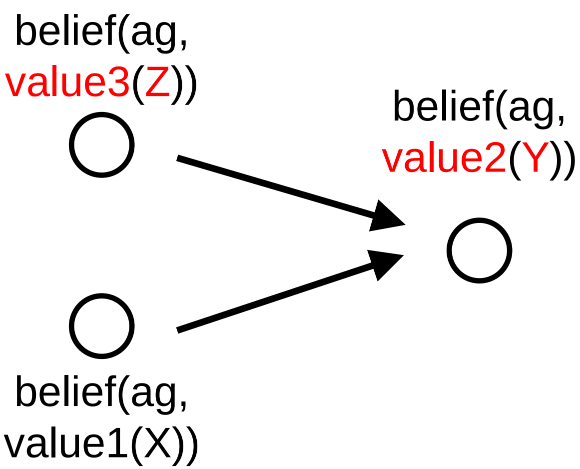
FORWARD

Numerical Values: Many-to-one



IF value1(X)
AND value2(Y)
THEN value3(X+Y)

DOMAIN



IF belief(ag, value3(Z))
AND belief(ag, value1(X))
THEN belief(ag, value2(Z-X))

BACKWARD

Overview of today's lecture

1. Summary previous lecture
2. Reasoning techniques
3. **Case study**

Case: VR Exposure Therapy

Case study

- Idea:
  - system that provides VR therapy
  - presents “scary” stimulus
  - however: anxiety should **not** become too high
- Analysis model:
  - assess when the user is (too) scared
- Support model:
  - generate action to reduce anxiety

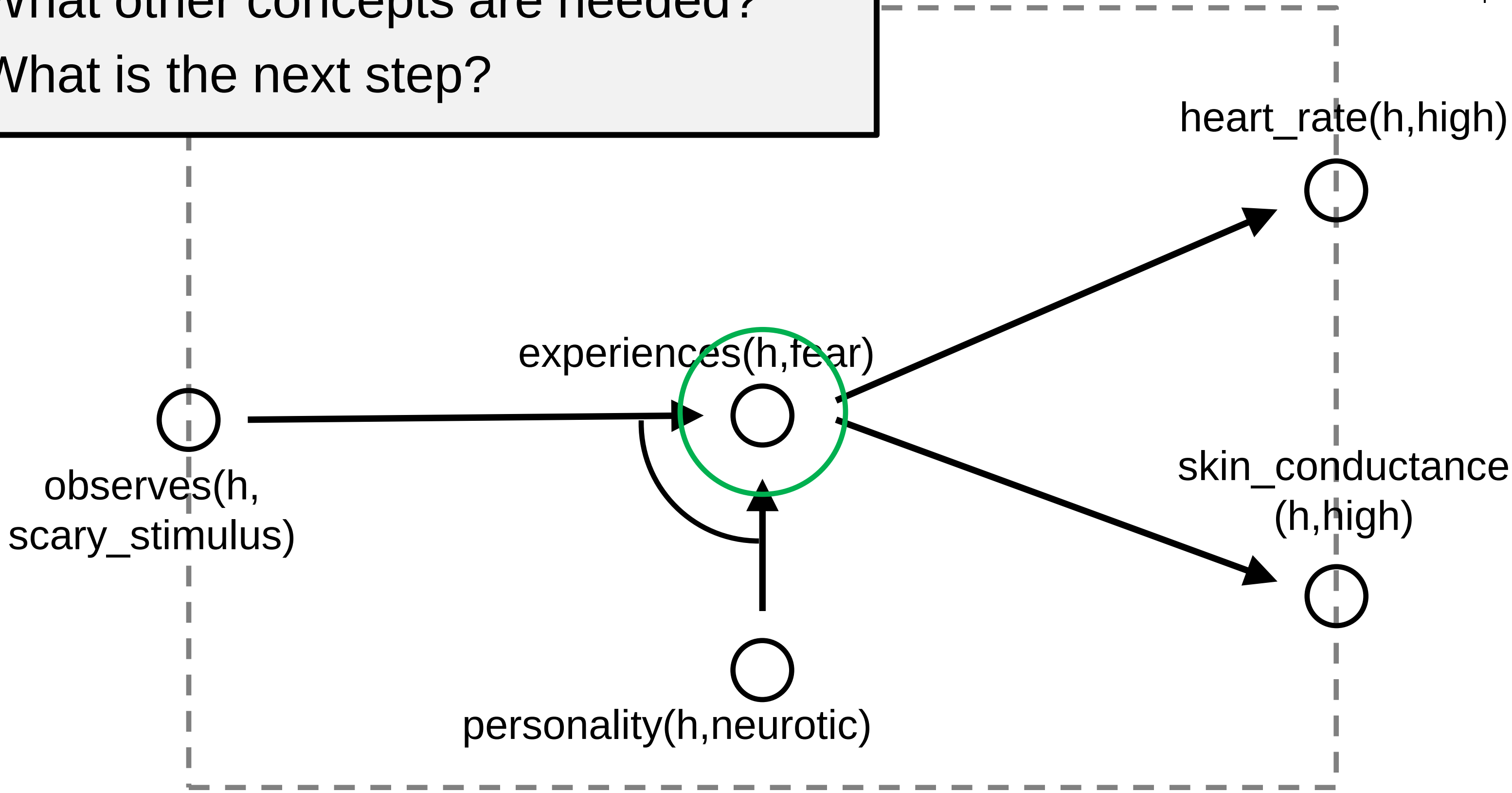
Analysis model - forward

- Assignment

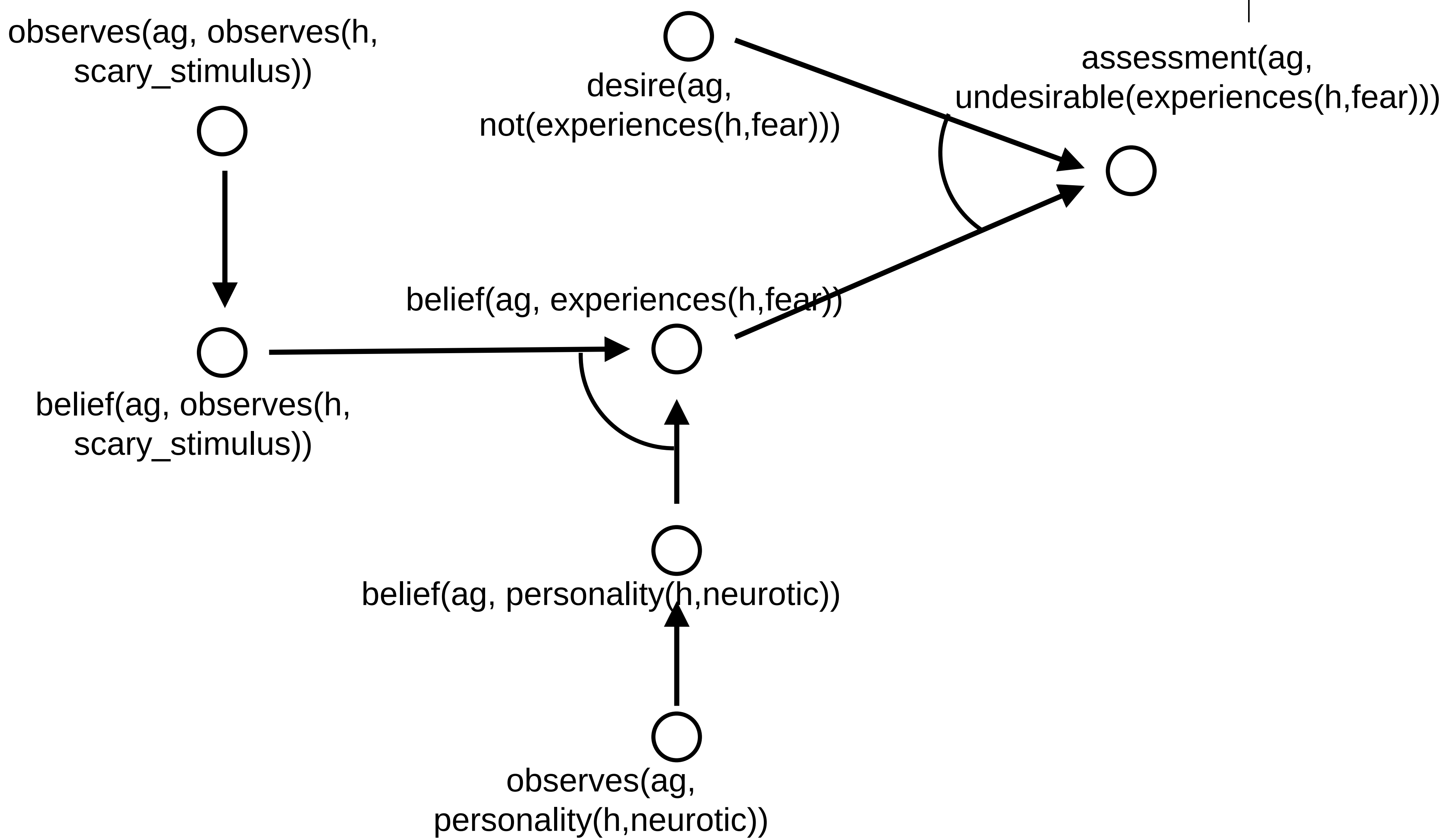
- Design an Analysis Model using forward reasoning for the domain model described on the next slide, with the aim to find out whether the anxiety became too high

- For which concept do we have a desire?
- For which concepts do we introduce observations?
- What other concepts are needed?
- What is the next step?

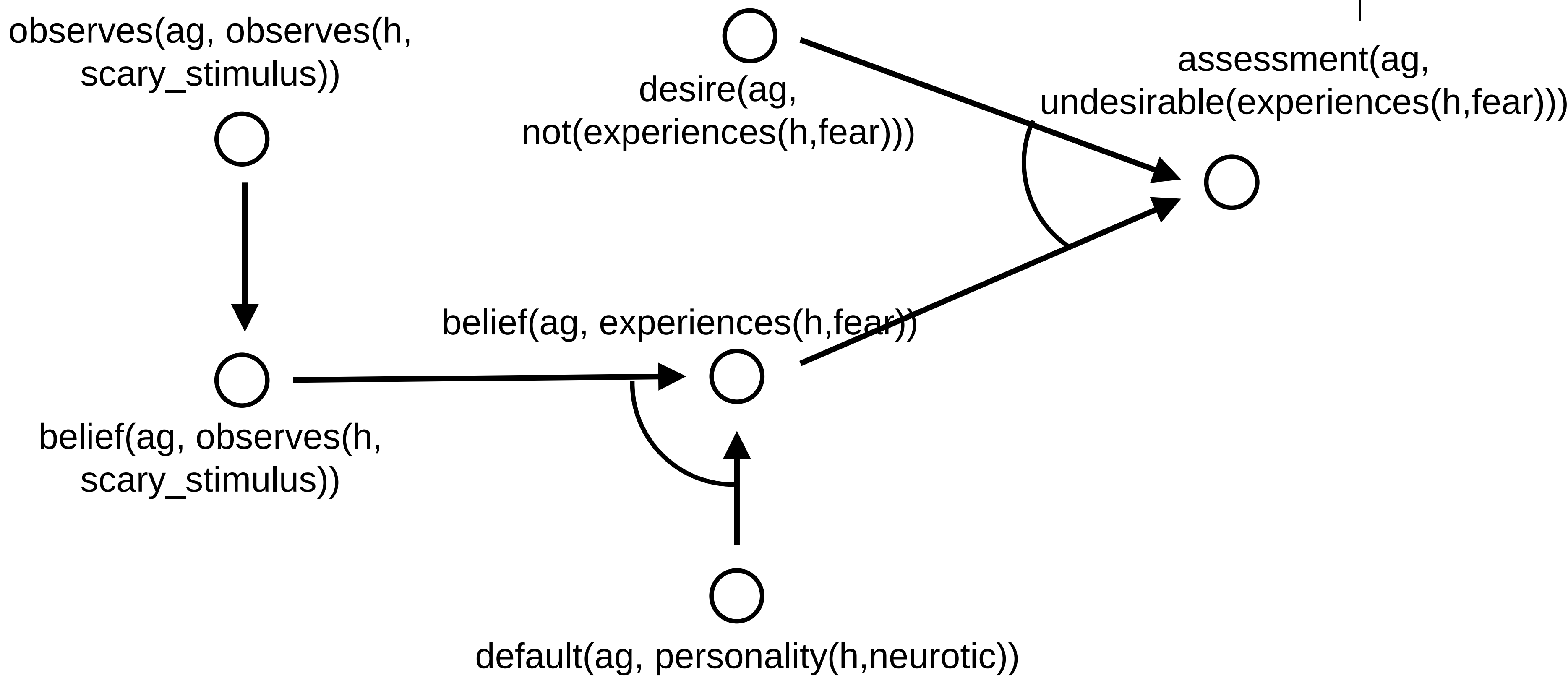
in model



Solution 1a: Forward



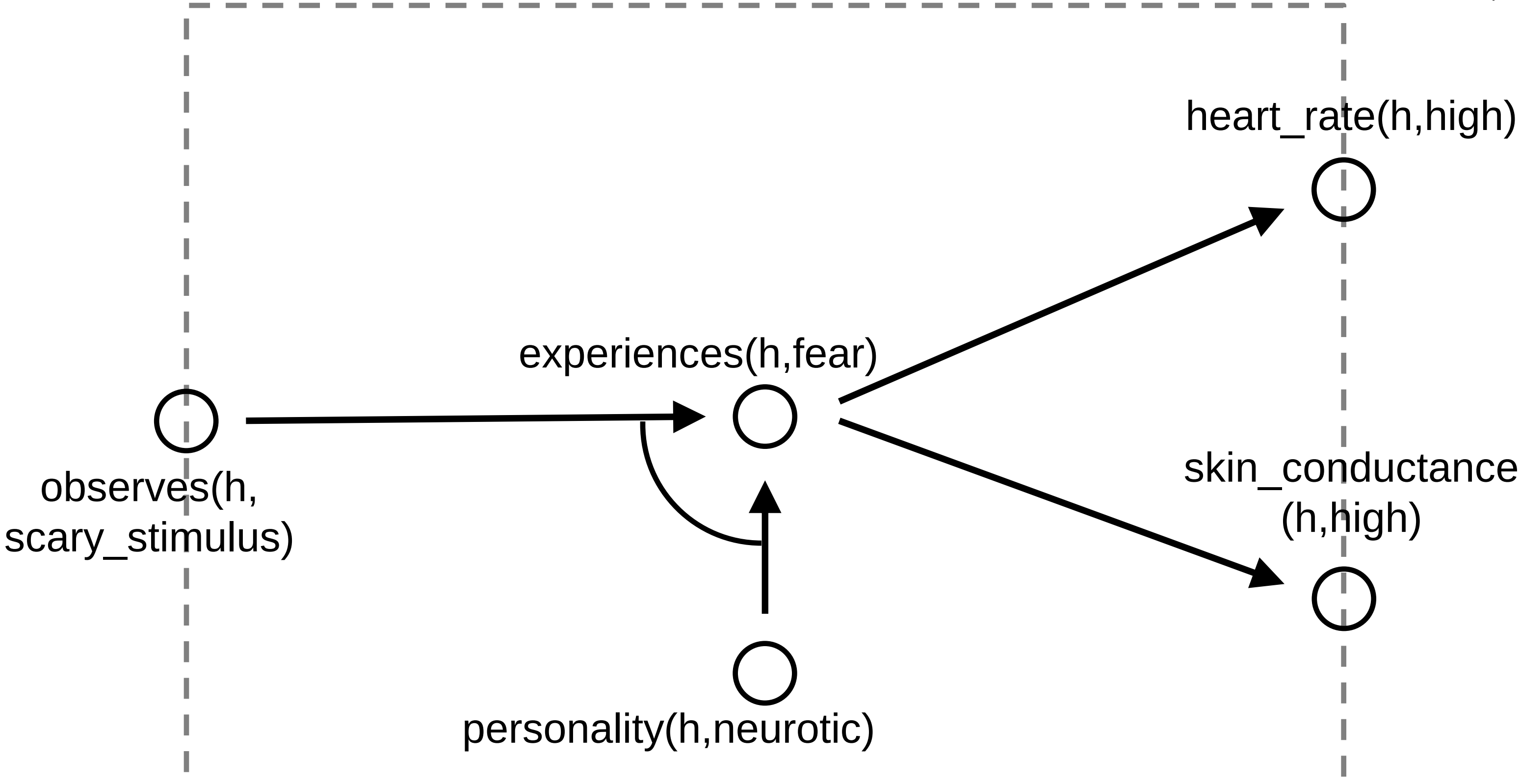
Solution 1b: Forward



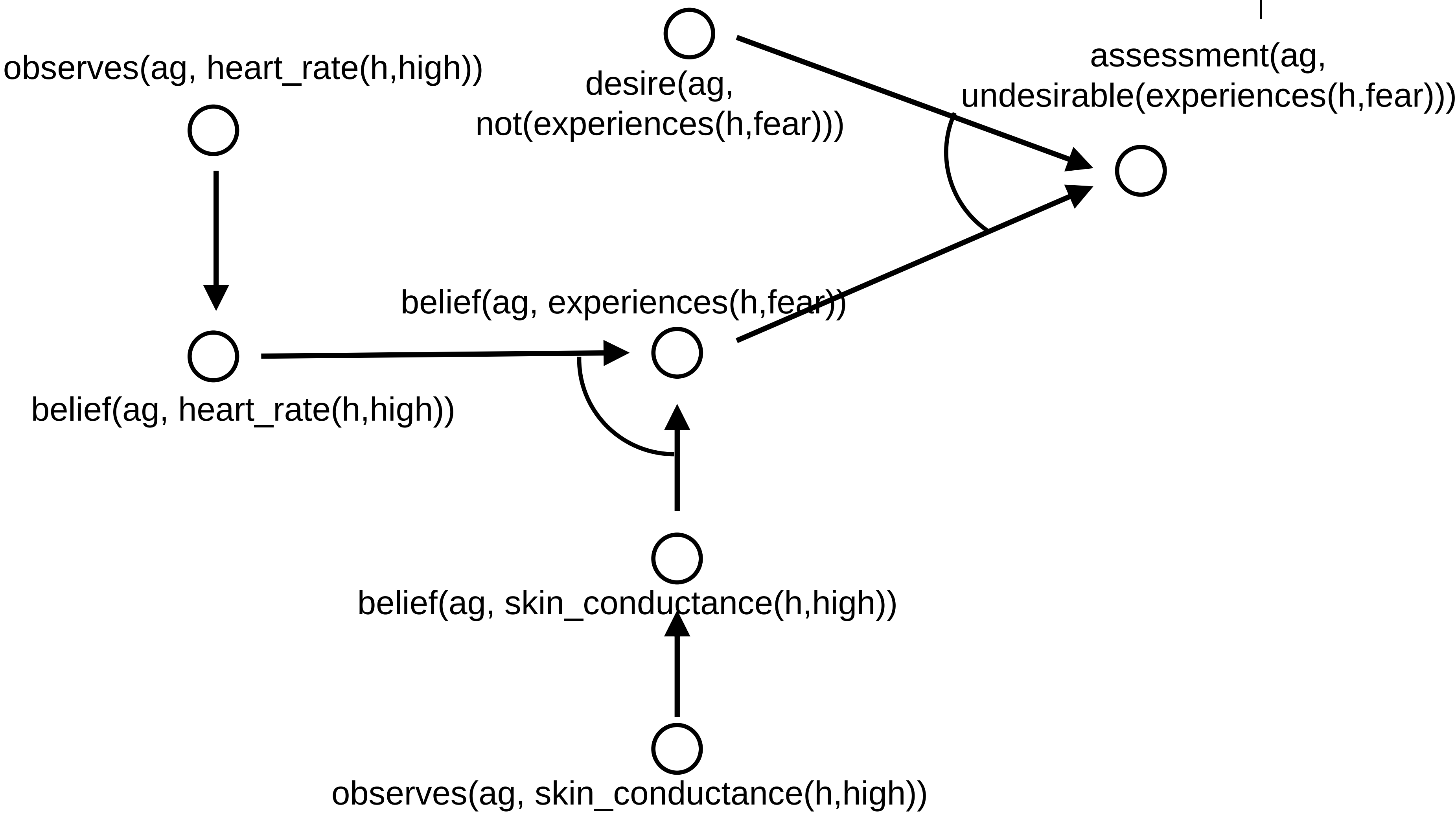
Analysis model - backward

- For which concepts do we introduce observations?
- What other concepts are needed?

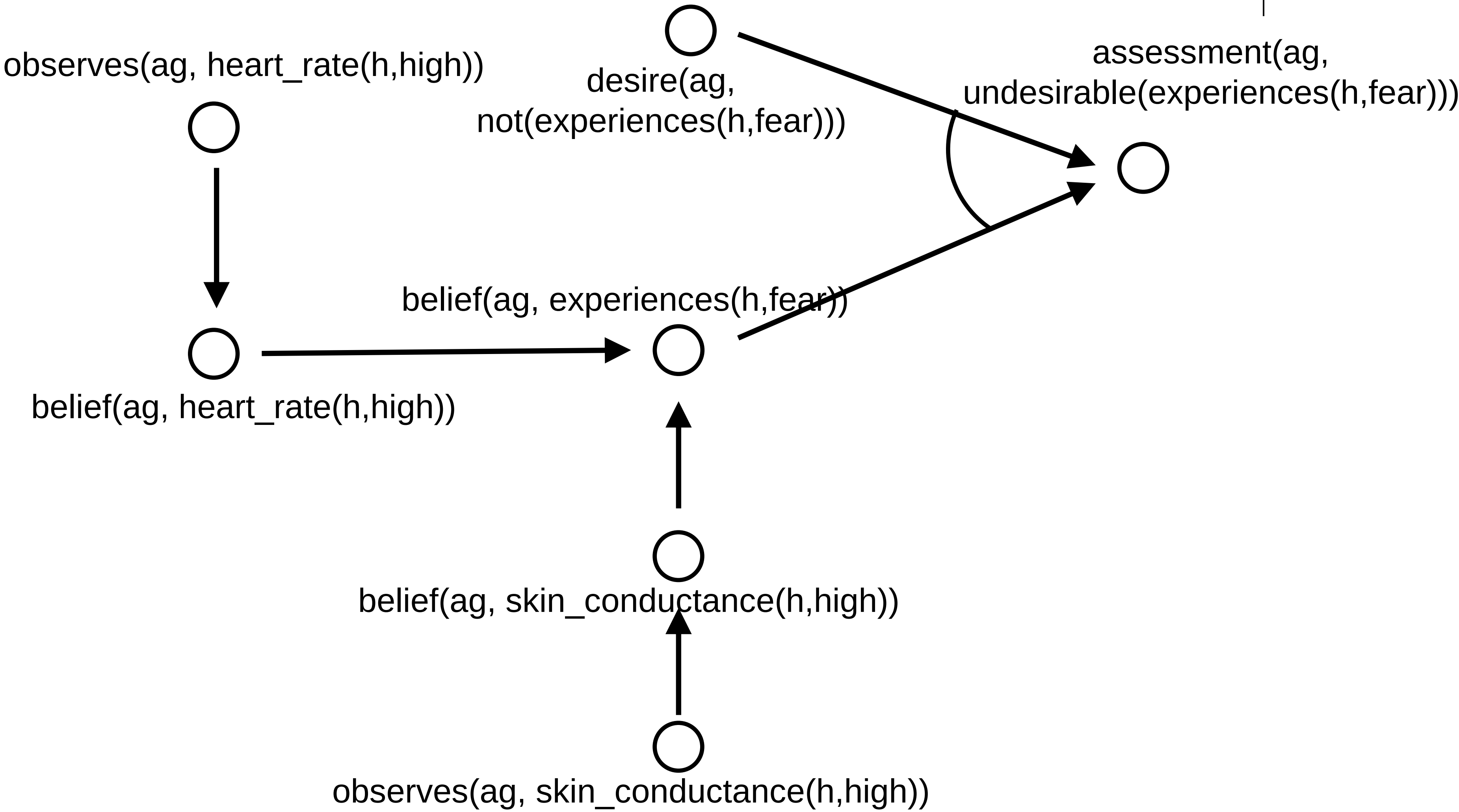
l Training



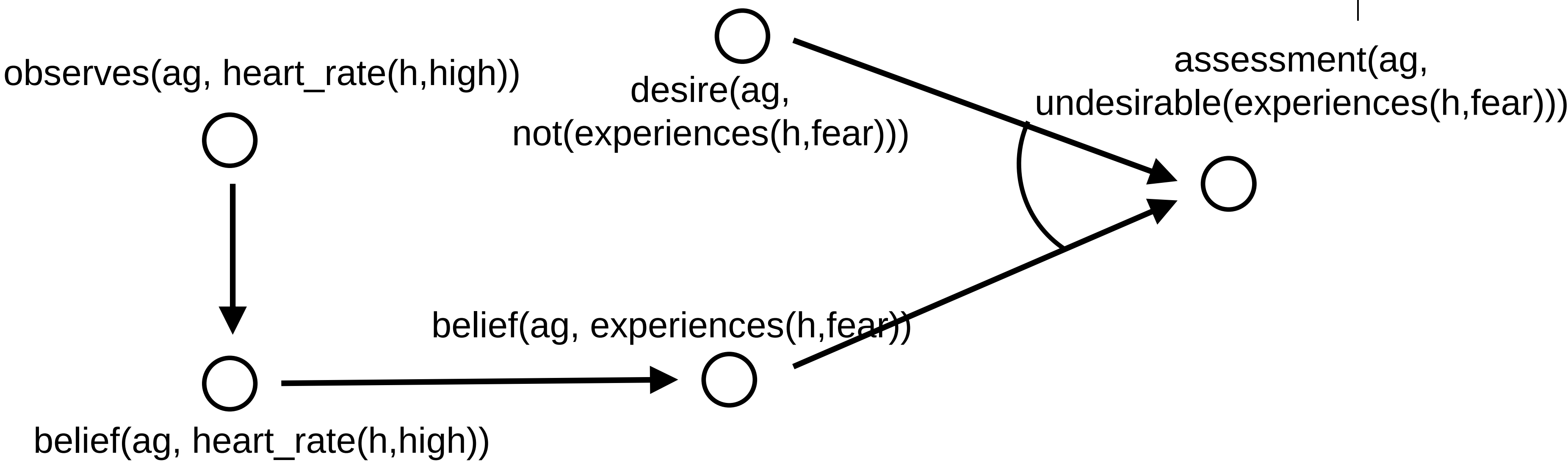
Solution 2a: Backward



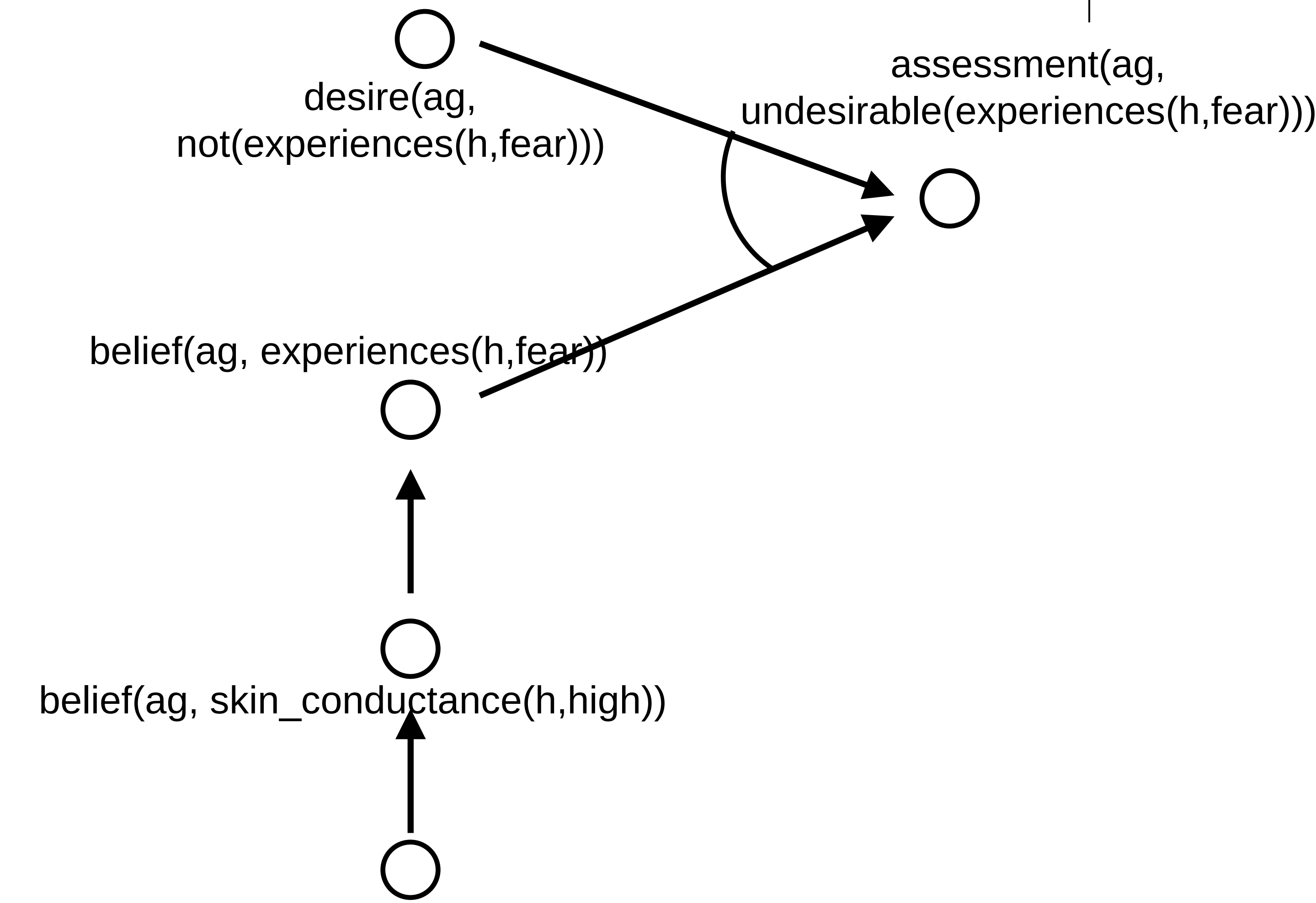
Solution 2b: Backward



Solution 2c: Backward



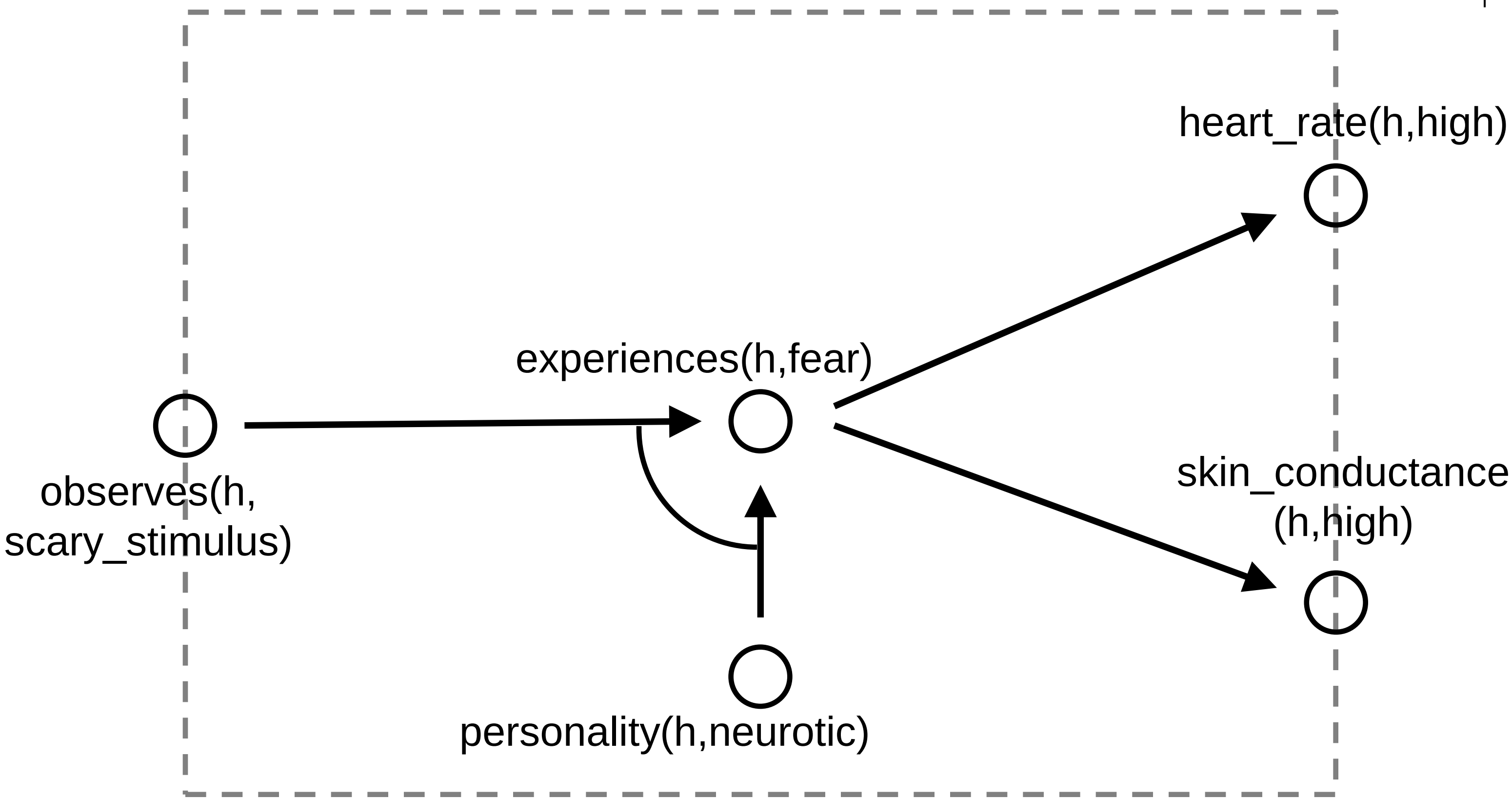
Solution 2d: Backward



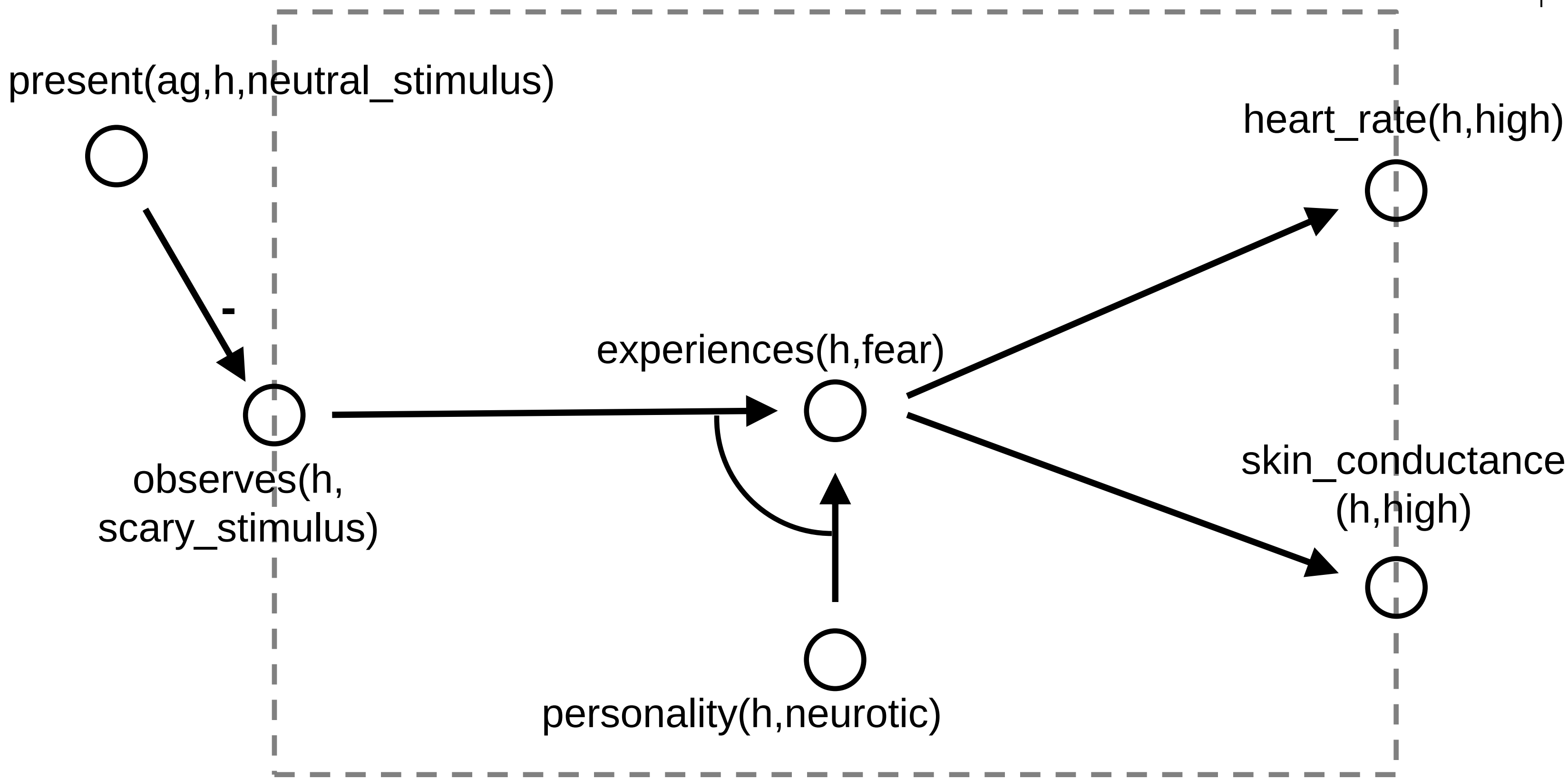
Support model

- First: add “support action”

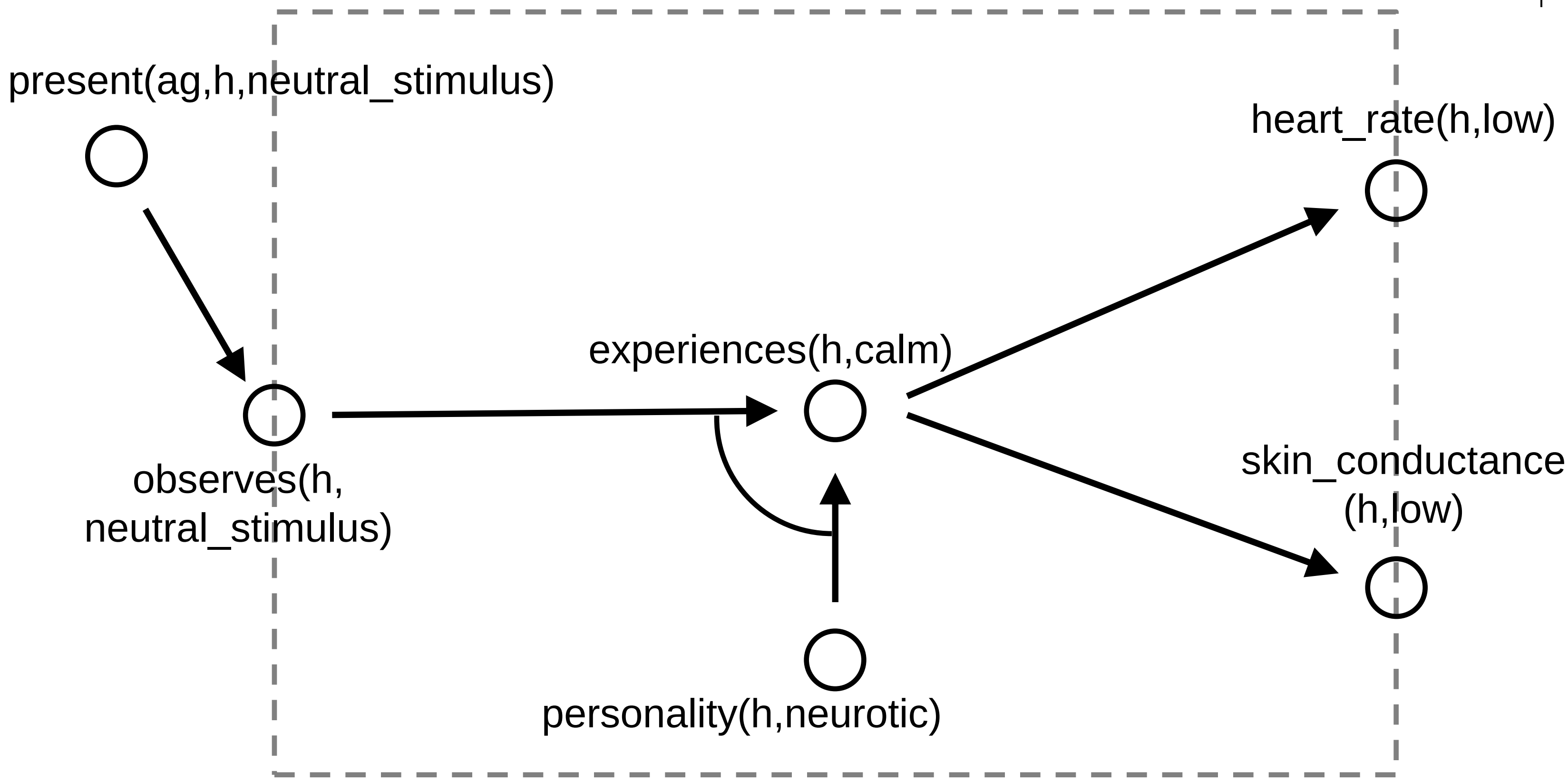
Case Study: Virtual Training



Case Study: Virtual Training

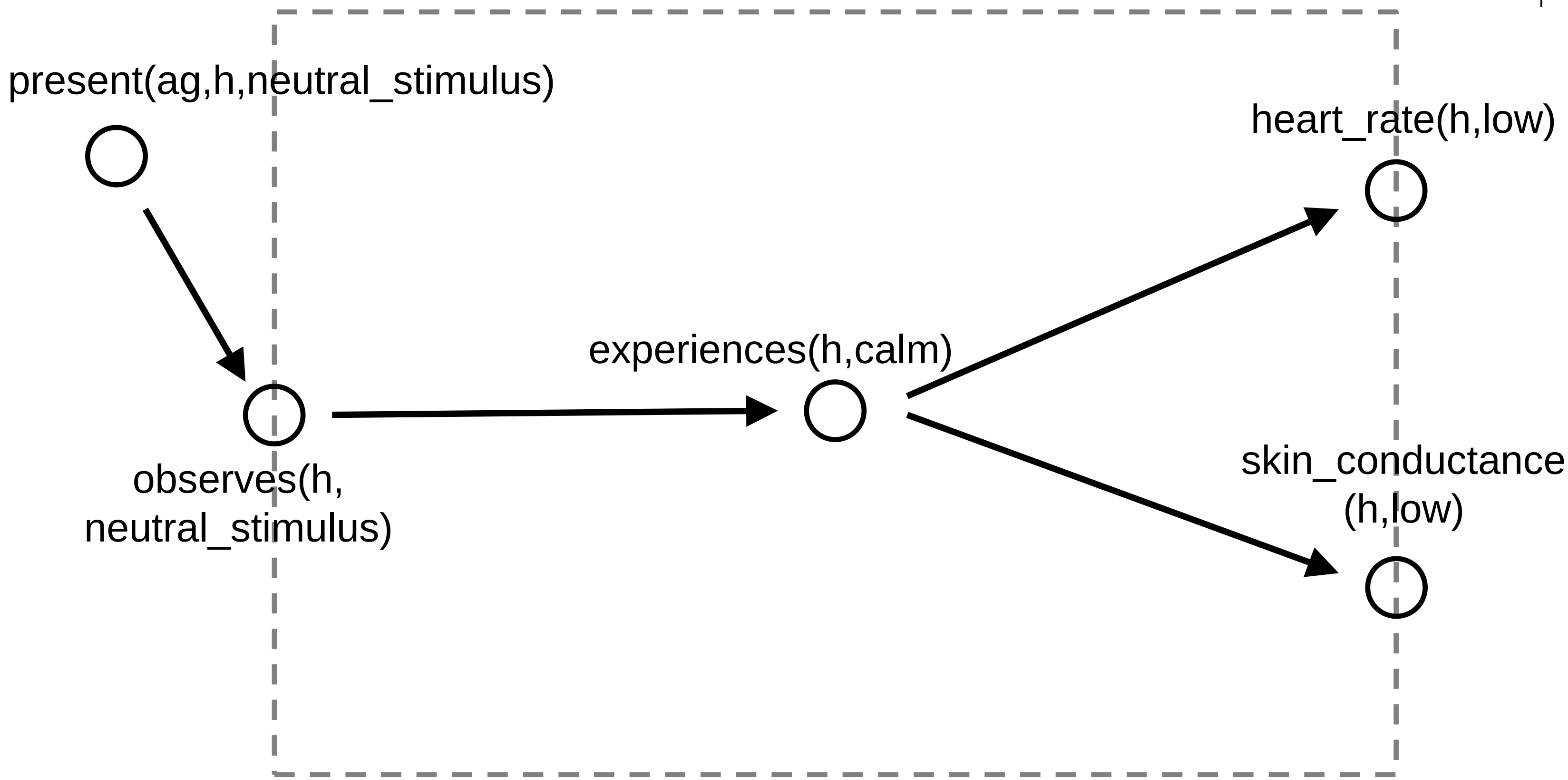


Case Study: Virtual Training



Propagate negative effects

Case Study: Virtual Training



Assume that personality is fixed.

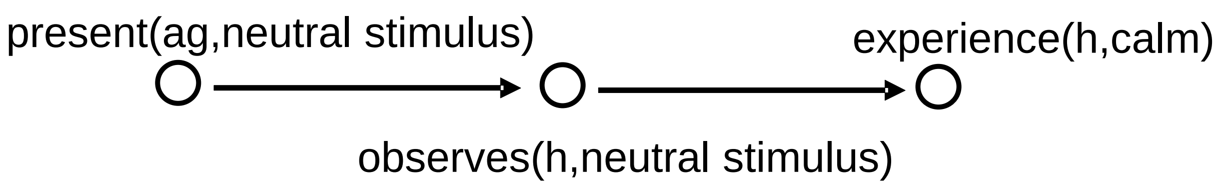
Assignment

- Design the **Support model** for the virtual training case study
- Both forward and backward!

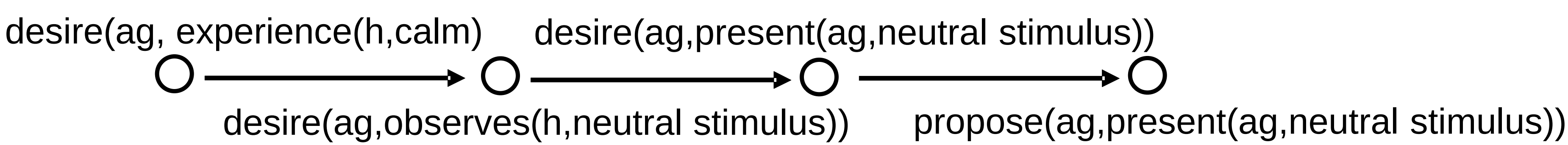
- What is the start of our reasoning?
- What is the next concept?
- What is the conclusion?

rd)

relevant part of the domain model

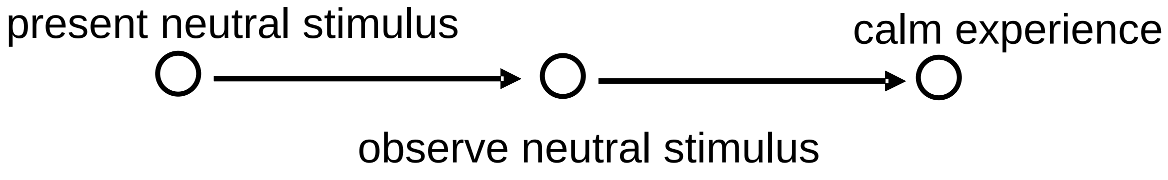


support model

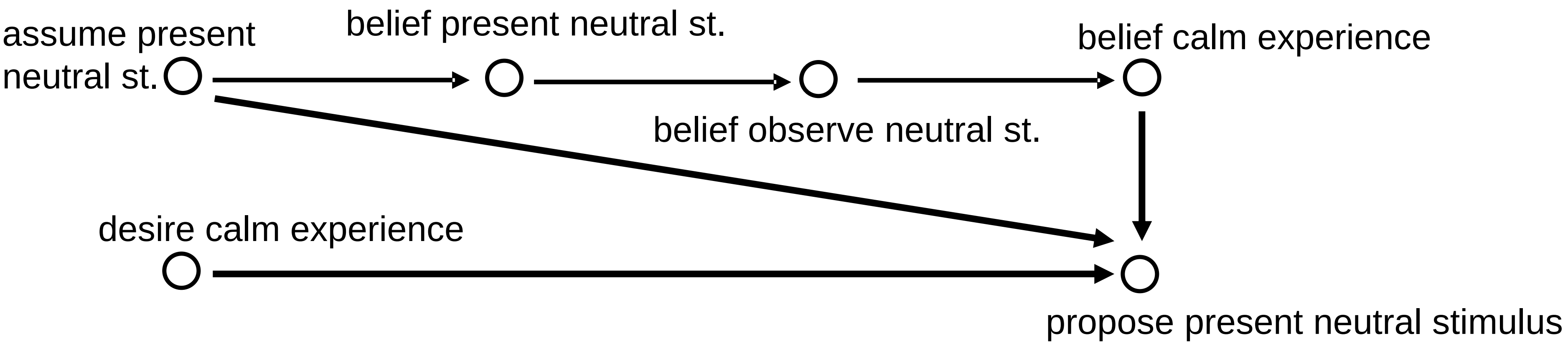


- What is the start of our reasoning?
- What is the next concept?
- And then?
- On which concepts does the conclusion depends?

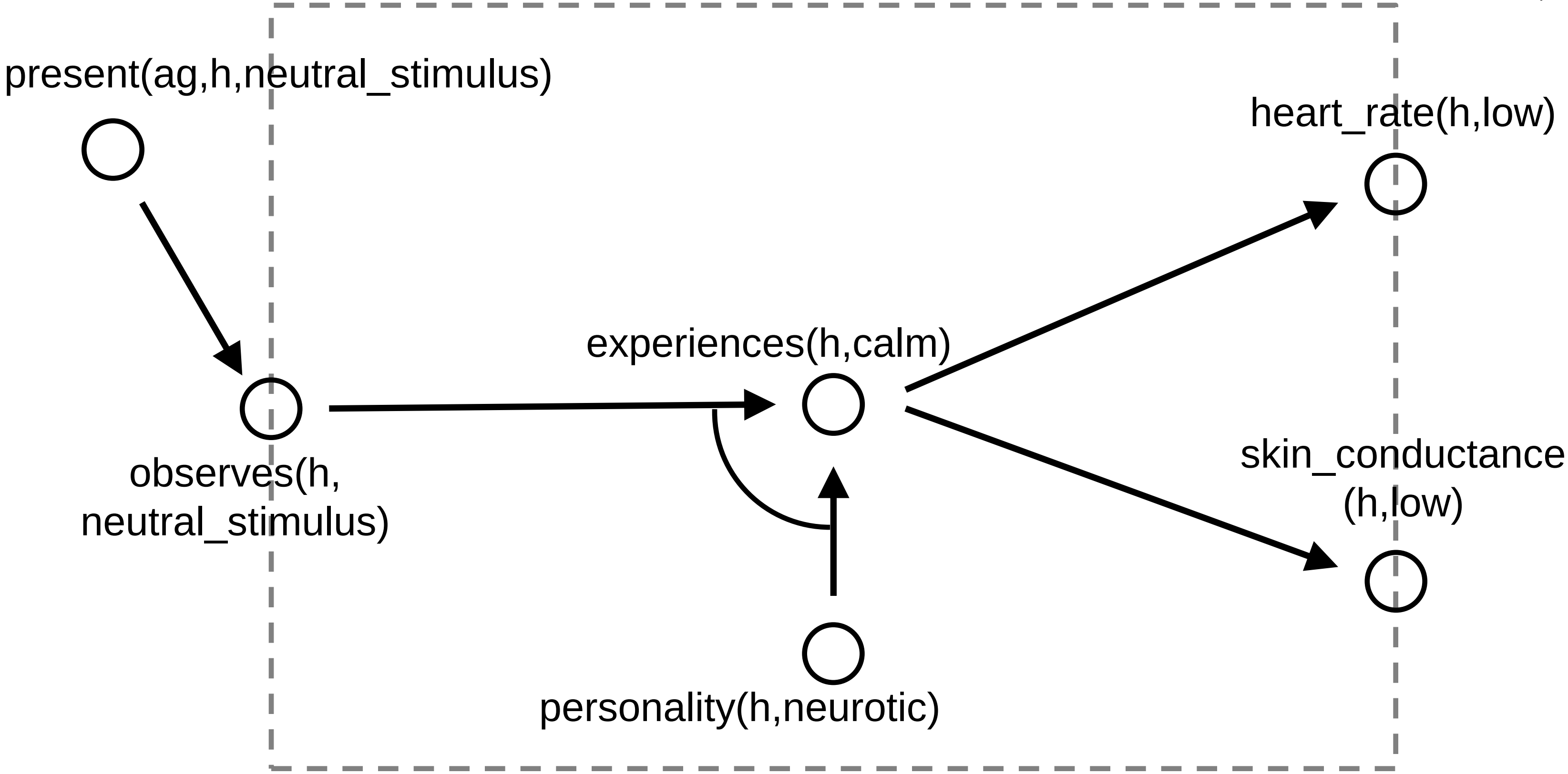
relevant part of the domain model



support model



Discussion



What if the neurotic personality is not taken as default?

Answer: **beliefs** from analysis model can be re-used!

Conclusion

- **Domain model → Analysis model**:
  - Use meta-statements about domain model
  - Reason from observations and desire to assessment
  - Forward or backward reasoning (or both)
- **Domain model → Support model**:
  - Extend domain model with support actions
  - Backward reasoning: from desired state to support action
    - Reasoning about **desires**
  - Forward reasoning: from **assumed** action to desired effect
    - Reasoning about (hypothetical) **beliefs**